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Combining Human-in-the-Loop Systems and AI Fairness Toolkits to Reduce Age Bias in AI Job Hiring Algorithms

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***Abstract*—As artificial intelligence (AI) systems become more sophisticated, they are increasingly integrated into high-stakes decision-making processes, such as hiring, fraud detection, loan approvals, and medical diagnoses. However, this growing reliance on AI raises concerns about the potential for these systems to perpetuate and amplify societal biases. Researchers have developed two main approaches to bias mitigation in AI to address this issue: human-in-the-loop (HITL) systems and AI fairness toolkits. HITL systems involve human reviewers actively participating in the AI decision-making process, while AI fairness toolkits are software tools that can identify and mitigate bias. HITL systems are particularly effective in addressing biases tied to specific domains, while AI fairness toolkits can be useful in identifying and addressing bias proactively. This paper examines different combinations of HITL systems and AI fairness toolkits, conducts an experiment to evaluate biases in hiring decisions using each, and provides recommendations for organizations considering implementing one or both approaches.**

Keywords—human-in-the-loop systems, artificial intelligence, AI fairness, AI bias mitigation, job hiring, artificial intelligence

I. Introduction

The use of AI algorithms in the hiring process holds the promise of addressing long-standing biases against underrepresented groups, such as people of color and women. By design, these algorithms aim to eliminate or significantly reduce biases held by human hiring managers, enabling merit-based decisions devoid of considerations related to race, gender, or protected status [1]. However, AI algorithms can inadvertently replicate biases due to their reliance on training data, as illustrated by Amazon's experience in 2015, where its recruiting tool exhibited gender biases in favor of male candidates [2].

While discussions about algorithmic biases have garnered substantial attention, age bias, or ageism, remains an underreported concern in the hiring process. As people live longer, governments have raised retirement ages and implemented measures to protect older workers from age-based termination. Nonetheless, several studies (e.g., [3-5]) have shown that age discrimination during hiring perpetuates the challenge of older job seekers securing new employment opportunities.

For several reasons, age bias is challenging for job search AI algorithms. First, AI algorithms often rely on historical data, which can contain inherent biases and favor past hiring practices that may have discriminated against older candidates. Second, AI systems might inadvertently prioritize keywords, skills, or attributes associated with younger applicants, disregarding the valuable experience and expertise that more senior candidates bring. Third, algorithms can perpetuate stereotypes about older workers, such as being less adaptable or tech-savvy, leading to biased recommendations. Lastly, the lack of diversity in training data can result in the underrepresentation of older individuals, making it difficult for AI to make unbiased recommendations for this demographic, exacerbating the age bias challenge in job searches. To tackle this pressing issue, researchers have developed two primary strategies for mitigating bias in AI: human-in-the-loop (HITL) systems and AI fairness toolkits. These two approaches represent distinct but complementary methods for addressing bias in AI applications.

HITL systems introduce human reviewers into AI decision-making, providing crucial oversight and expertise. These reviewers can identify nuances, contextual intricacies, and domain-specific biases that AI may miss, particularly in industry-specific settings. HITL effectively addresses biases tied to specific domains, enhancing fairness and accuracy by combining human and AI capabilities.

On the other hand, AI fairness toolkits are software tools designed to detect and mitigate bias within AI systems proactively. These toolkits analyze algorithms, datasets, and decision processes to uncover subtle or overt biases. Once identified, they offer mechanisms to rectify or mitigate these biases, promoting fairness and equity. AI fairness toolkits are versatile and can be applied to a wide range of AI applications, making them valuable in addressing bias on a broader scale, not limited to domain-specific concerns.

While HITL systems excel in addressing domain-specific biases through human intervention and expertise, AI fairness toolkits provide a systematic and scalable approach to identifying and rectifying bias in AI systems. Together, these two approaches form a robust strategy for mitigating bias in AI, offering the potential to create more equitable and fair AI applications while harnessing the power of artificial

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intelligence in high-stakes decision-making contexts. This ongoing effort to strike a balance between the benefits of AI and the need for fairness is a critical step in ensuring that AI serves as a force for positive change in our increasingly automated world.

To eliminate age biases in job searches, HITL systems engage human reviewers in actively assessing job candidates to ensure impartial and equitable selection. Ideally, these reviewers consider applicants' qualifications and experience free from age-based discrimination. AI fairness toolkits complement this process by continuously monitoring and analyzing the algorithms used in job searches, identifying age-related biases, and providing recommendations to the AI system to prevent age discrimination in the recruitment process. These two strategies work collaboratively to foster fair hiring practices, irrespective of an applicant's age, and enhance the overall equity of the job search process.

This paper examines different strategies involving HITL systems and AI fairness toolkits by experimenting to examine age biases and then examine mitigation strategies. The remainder of the paper is organized as follows. In the next section, we discuss the background and motivation of age discrimination and how HITL and AI Fairness Toolkits can help mitigate age bias issues in AI algorithms. In Section 3, we provide related work, and in Section 4, we describe our experiments. Next, we discuss our results in Section 5, and last, we conclude our work in Section 6.

II. Background and Motivation

A. *Age Bias in the Hiring Process*

Age bias in the hiring process poses a significant problem with multiple adverse consequences for individuals, organizations, and society. Harris [6] found that age bias is more difficult to remove from AI algorithms using AI fairness toolkits than race or gender in job hiring results.

Age bias results in unfair and discriminatory hiring decisions, as employers deny opportunities to older workers based on age rather than qualifications or abilities. This bias impedes older workers from finding suitable employment and hinders their ability to provide for themselves and their families [7]. Furthermore, age bias hampers diversity and innovation within the workforce by excluding a valuable pool of experienced talent. This lack of diversity can stifle innovation and impede organizations' adaptability to change, affecting their long-term success [8]. Age bias also has economic ramifications, diminishing productivity and growth, increasing healthcare costs, and impacting social security payouts [9].

Another concern is that age bias contributes to ageism within society, normalizing discriminatory practices in the hiring process and reinforcing negative stereotypes about older individuals. Age bias can make it harder for older individuals to receive the respect and dignity they deserve, ultimately undermining societal values [10]. Beyond these effects, age bias can damage the morale of older workers already in the workforce, leading to feelings of undervaluation and unappreciation, resulting in decreased productivity and higher turnover rates. Additionally, it can discourage older individuals from pursuing further education and training, limiting their earning potential and their competitiveness in the job market [3]. Age bias perpetuates a cycle of inequality, making it more challenging for older workers to escape poverty and disadvantage.

The terms age bias and age discrimination are often used interchangeably, but subtle distinctions exist between them. *Age bias* refers to negative attitudes or beliefs about people based on age. These attitudes can be conscious or unconscious, leading to stereotypes and prejudices about older adults [11]. *Age discrimination* is the act of treating someone differently based on their age. This discrimination can happen in various contexts, such as employment, housing, healthcare, and other aspects of life [12]. Age discrimination is often illegal and can have serious consequences for the individual being discriminated against. This paper focuses on age bias, which is more implicit and more of a challenge for AI algorithms to eliminate [6].

B. *HITL AI*

HITL AI is a collaborative approach that leverages both artificial intelligence (AI) and human intelligence for superior outcomes. AI handles most tasks in this paradigm, while human experts intervene when necessary. HITL is ideal for complex, nuanced tasks like medical diagnosis, fraud detection, and customer service, where AI alone may fall short.

HITL harnesses the strengths of both AI and human intelligence. AI excels at processing data and recognizing patterns but struggles with common sense, empathy, and ambiguity. Humans excel in these areas, making them valuable partners in AI development. HITL offers several advantages:

- *Enhanced Accuracy and Reliability*: Human oversight corrects AI errors, ensuring more trustworthy results.
- *Increased Transparency and Explainability*: Human involvement makes decision-making more understandable and helps identify biases.
- *Responsible and Ethical AI Development*: HITL ensures AI systems align with human values and ethical principles.

HITL AI is employed in domains like medical diagnosis, fraud detection, customer service, content moderation, and autonomous vehicles. With the progression of AI technology, HITL AI is positioned to exert an even more substantial influence.

C. *AI Fairness Tooklits*

AI fairness toolkits are software tools that can identify and mitigate bias in AI systems. These toolkits can provide insights into AI decision-making, identify potential biases, and suggest corrective measures. AI fairness toolkits can be useful for organizations that want to identify and address bias in their AI systems proactively.

Two of the most popular are IBM's AI Fairness 360 Toolkit [13] and Microsoft FairLearn Toolkit [14]. These two open-source toolkits help developers identify and mitigate bias in machine learning models. While they share a common goal,

TABLE I. KEY DIFFERENCES BETWEEN MAJOR AI FAIRNESS TOOKLITS

Feature	IBM AI Fairness 360	Microsoft FairLearn
Strengths	Wide range of metrics, ability to detect and mitigate bias in a variety of models, extensible framework	Easy to use, well-documented, focuses on fairness through awareness, and includes a Jupyter widget for model assessment.
Weaknesses	Complex to use, not as well-documented	A smaller range of metrics primarily focused on classification models
Overall	More powerful and versatile	More user-friendly and focused

they have different strengths and weaknesses. Overall, IBM AI Fairness 360 is a more powerful and versatile toolkit, but Microsoft FairLearn is a good choice for users who want a more user-friendly and focused tool [15]. Table 1 provides an overview of the relative strengths and weaknesses of each.

III. RELATED WORK

HITL has emerged as a promising strategy to combat AI biases in the context of hiring. Multiple research studies have delved into the effectiveness of HITL in mitigating bias and enhancing fairness throughout the hiring process.

A. HITL to Address AI Biases in Hiring

HITL has emerged as a promising strategy to combat AI biases in the context of hiring. Multiple research studies have delved into the effectiveness of HITL in mitigating bias and enhancing fairness throughout the hiring process. In a study by Dastin et al. [16], HITL was applied to reduce bias in resume screening. The researchers developed an AI system to identify potentially biased language in resumes and then flagged such cases for human review. Their findings demonstrated that HITL significantly decreased bias in the hiring process, resulting in a more equitable selection of candidates.

Similarly, in a simulated experiment by Zhang et al. [17], the HITL process enhanced the fairness and accuracy of hiring decisions compared to exclusively AI-driven or human-driven methods. Kim et al. [18] introduced a framework for incorporating human feedback into AI-driven hiring decisions, revealing that HITL AI notably reduced bias and improved fairness, particularly for underrepresented groups. Yang et al. [19] focused on applying HITL to identify and correct biases in AI-generated hiring recommendations. Their study showed that HITL effectively recognized and rectified biased recommendations, leading to a more equitable candidate selection process.

In a comprehensive evaluation of HITL AI for bias mitigation in hiring, Chen [20] analyzed various HITL strategies and their effectiveness in reducing bias and enhancing fairness. Their results indicated that HITL AI can be a valuable tool for promoting equity in the hiring process. Lastly, Park [5] investigated the influence of HITL on age bias in AI hiring decisions, revealing that HITL can contribute to reducing age bias. However, it also highlighted the need for further research to determine the long-term effects of HITL on age bias.

B. Comparing HITL with AI Fairness Toolkits to Address Job-Related AI Biases

Several studies have investigated the effectiveness of HITL systems and AI fairness toolkits in mitigating age bias in hiring. Chen et al. [21] comprehensively evaluated various HITL strategies and AI fairness toolkits, finding that both approaches can be effective. Still, the most effective approach depends on the specific context and resources of the organization. Zhang et al. [22] developed a HITL AI system that identified potentially biased language in resumes and flagged them for human review, significantly reducing age bias in the hiring process. Harris [6] highlighted the challenges of mitigating age bias in AI-driven hiring processes, noting that age biases were more difficult for AI fairness tools to address than gender and race biases. Neumark et al. [7] proposed combining data preprocessing and model regularization techniques to mitigate age bias effectively in resume-screening AI models. Zhang and Chen [23] proposed a HITL approach to fairness monitoring for machine learning systems, including hiring systems, suggesting that HITL monitoring could be more effective than traditional automated methods in detecting and mitigating age bias. Kim et al. [24] compared HITL systems and AI fairness toolkits, finding that HITL systems were more effective in mitigating age bias. However, AI fairness toolkits could still be useful for identifying potential biases.

IV. EXPERIMENT METHODOLOGY

We examine different processes as illustrated in Fig. 1. We evaluated several AI Fairness Toolkit methods, a HITL approach, and a combination of these to see which solutions best mitigate age biases in selecting candidates for job interviews.

A. Data Collection and Preprocessing

We obtained a job listing for a mid-level IT position randomly selected from a dataset of 50,000 online job listings on Reed UK. We limited our search to full-time positions with an annual salary range in greater London, UK. We then obtained candidate CVs by scraping individual CVs from the livecareer.com website. We randomly selected 800 CVs that indicated an objective matching the job posting obtained from Reed UK and where the candidate's age could be reasonably determined. These CVs regarding race, gender, and age were anonymized to the best of our ability while ensuring that essential job-related information about the candidates remained intact. We kept track of the candidates' ages separately but did not disclose this to the experts.

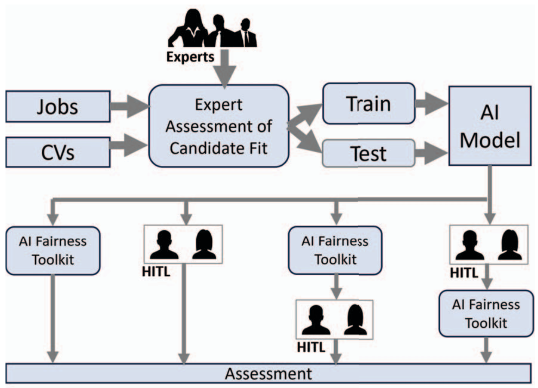

Fig 1. Process flow for our experiment, showing the four types of evaluations made.

Three executive recruiters, each possessing an average of 13.5 years of experience in IT job recruitment, individually evaluated all 800 CVs. Their task was to make binary (yes/no) selections for initial interviews relevant to the specified job position. In a real-world application, the collective expertise of these professionals, combined with the CV data, would serve as inputs for an AI algorithm. Utilizing this dataset, we subsequently developed an AI algorithm to make binary (yes/no) determinations for initial interviews in relation to the specified job role. Our training dataset encompassed 400 randomly selected CVs, while our test set included 200. The last 200 CVs were reserved to determine the effectiveness of retraining our model after HITL or AI Fairness Toolkit changes (described later in this section as our second test set).

B. Bias Evaluation Metrics

We examined the following three metrics to determine algorithmic bias:

1) Statistical Parity Difference (SPD)

SPD is a measure of fairness that compares the probability of a positive outcome for two groups, typically based on a protected attribute such as race or gender. SPD is calculated as the absolute difference between the probability of a positive outcome for the two groups. A value of 0 indicates no difference in the probability of a positive outcome for the two groups. In contrast, a positive value indicates that the protected group (i.e., older workers, which we defined as 50 or older) is less likely to receive a positive outcome. A negative value indicates that the protected group is more likely to receive a positive outcome.

$$SPD = | P(Y=1 \mid D=\text{group 1}) - P(Y=1 \mid D=\text{group 2}) |$$

where:

$P(Y=1 \mid D=\text{group 1})$ is the probability of a positive outcome for the protected group

$P(Y=1 \mid D=\text{group 2})$ is the probability of a positive outcome for the unprotected group

2) Disparate Impact (DI)

DI is a measure of fairness that compares the positive outcome rate for two groups, typically based on a protected attribute. DI is calculated as the ratio of the rate of a positive outcome for the protected group (older workers) to the rate of a positive outcome for the unprotected group. A value of 1 indicates that the two groups have the same positive outcome rate. In contrast, a value greater than 1 indicates that the protected group is more likely to receive a positive outcome. A value less than 1 indicates that the protected group is less likely to receive a positive outcome.

$$DI = P(Y=1 \mid D=\text{group 1}) / P(Y=1 \mid D=\text{group 2})$$

where:

$P(Y=1 \mid D=\text{group 1})$ is the probability of a positive outcome for the protected group

$P(Y=1 \mid D=\text{group 2})$ is the probability of a positive outcome for the unprotected group

3) Theil Index (T)

T, a measure of inequality, is calculated as the weighted sum of the squared differences between the probability of a positive outcome for each individual and the average probability of a positive outcome. A value of 0 indicates perfect fairness, while a higher value indicates greater inequality.

$$T = \Sigma\, i\, w_i (p_i - \bar{p})^2$$

where:

w_i is the weight for an individual i

p_i is the probability of a positive outcome for individual i

$\bar{p}$ is the average probability of a positive outcome.

In the context of fairness in machine learning, the weights, w_i, can be assigned based on the protected attribute (in our case, age - which is the average opportunity for older adults being asked to an interview across disciplines in the US, or 26.24% as of 2023[1]). This assignment of weights allows the Theil Index to assess whether the probability of a positive outcome is fair across different groups.

C. AI Fairness Toolkit Only Treatment

The AI Fairness 360 Toolkit has several different methods to reduce bias. We performed a pilot study to determine which AI Fairness 360 toolkit methods should be applied to our algorithm and evaluated the resulting outputs. This toolkit provides several techniques that involve pre-processing (modifying the training data), in-processing (changing the learning algorithm), and post-processing (changing the predictions). Fig. 2 provides a data flow of these methods. Fig 1 provides a data flow of different techniques from the AI Fairness 360 Toolkit we used in this study.

[1] https://www.zippia.com/advice/job-interview-statistics/

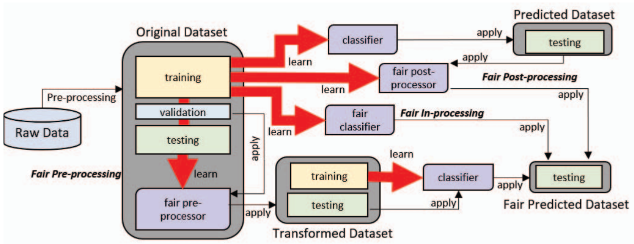

Fig 2. Data flow of pre-processing, in-processing, and post-processing methods of the AI Fairness 360 Toolkit. Adapted with permission from [15].

From Fig. 2, we see numerous approaches are available within the toolkit. Harris [25] evaluated the AI Fairness 360 toolkit. He identified the most effective techniques within the toolkit for HR assessment, which included disparate impact reduction for pre-processing, adversarial debiasing for in-processing, and equalized odds post-processing. Combining these three techniques achieved the best results (see Table 2); consequently, we implemented this same combined approach in our study. For a more comprehensive understanding of these fairness metrics, refer to [13] and [26]. Note that numbers in italics in Table 2 denote fairness based on the definitions outlined in [13]. We perform our statistical tests on Disparate Impact (DI) since this is the most widely used factor in the literature of the three metrics we examined.

Table 2 illustrates the change in selection percentage, the Statistical Parity Difference, the Disparate Index, and the Theil index for younger (Y) and older (O) workers. We found no interactions between the three factors; however, there was a statistically significant difference between ages as determined by one-way ANOVA ($F(3,396) = 9.261$, $p < 0.001$). We can see that the combination of pre-, in-, and post-processing results in the best combination and the best bias correction.

D. *HITL System-Only Treatment*

We wanted to assess the fairness of using humans-in-the-loop. To do so, we hired 96 crowdworkers from Amazon Mechanical Turk to examine the CVs and the ratings given as outputs from the AI model and determine if there were any age-based biases. Each crowdworker was given the AI algorithm's decision and the anonymized CV (but not the candidate's age) and was asked to determine whether the algorithm had made the correct decision. To do so, we had each candidate evaluate a pool of 50 candidate CVs. We tried two approaches – one in which the crowdworker was given preliminary training on age biases before assessment and one in which they were not. We also used two crowdsourcing models – one in which a single crowdworker determined if the algorithm's decision was fair and another in which we took the majority opinion of three crowdworkers (like the approach with the experts). The crowdworkers' corrections were made to the first test set for the model. The first test set of 200 CVs and the original training set of 400 candidate CVs were fed into our AI algorithm, and a decision was made on the second test set of 200 candidate CVs. The results are given in Table 3.

TABLE II. REDUCTION IN BIAS ON AGE BY APPLIED AI FAIRNESS 360 TECHNIQUE

Method Applied	Change in Selection Pct	SPD	DI	T
None	Y = 0.758, O = 0.525, **|Δ| = 0.000**	0.308	1.44	0.33
A. disparate impact reduction (pre-processing)	Y = 0.692, O = 0.591, **|Δ| = 0.067**	0.145	*1.18*	0.21
B. adversarial debiasing (in-processing)	Y = 0.702, O = 0.581, **|Δ| = 0.057**	0.169	*1.21*	0.15
C. equalized odds (post-processing)	Y = 0.684, O = 0.600, **|Δ| = 0.076**	0.124	1.39	0.28
Combination of methods A, B, and C	Y = 0.654 O = 0.629, **|Δ| = 0.105**	*0.041*	*1.05*	0.05

Y = younger (< 50 years of age), O = older (50+ years of age),
SPD – statistical parity difference, DI = disparate index, T = Theil index

From Table 3, we can observe that the approach with preliminary training worked best; however, the model in which a majority vote from 3 crowdworkers only marginally outperformed the model using the single crowdworker, although it required three times the cost (and time) to implement.

A two-way ANOVA was performed to analyze the effect of the *decision model* (single vs. majority vote) and *training approach* (no training vs. pre-training) on DI. It revealed that there was a statistically significant interaction between the effects of the decision model and training approach ($F(2, 596) = 5.092$, $p < 0.001$). Simple main effects analysis showed that the decision model did not significantly affect DI ($p = 0.3765$). However, the training approach did have a statistically significant effect on DI ($p < 0.000$).

E. *Evaluating the AI Fairness Toolkit - HITL System Order*

TABLE III. REDUCTION IN BIAS ON AGE BY APPLIED HITL TECHNIQUE

Method Applied	Change in Selection Pct	SPD	DI	T
None	Y = 0.758, O = 0.525, **|Δ| = 0.000**	0.308	1.44	0.33
Untrained Crowdworkers (N= 12) Single Crowdworker Makes Decision	Y = 0.744, O = 0.542, **|Δ| = 0.056**	0.166	1.24	0.22
Untrained Crowdworkers (N= 36) Majority Vote of 3 Crowdworkers Makes Decision	Y = 0.737, O = 0.545, **|Δ| = 0.054**	0.165	1.23	0.21
Pre-Trained Crowdworkers (N= 12) Single Crowdworker Makes Decision	Y = 0.679, O = 0.606, **|Δ| = 0.087**	*0.083*	*1.10*	0.08
Pre-Trained Crowdworkers (N= 36) Majority Vote of 3 Crowdworkers Makes Decision	Y = 0.691, O = 0.610, **|Δ| = 0.089**	*0.082*	*1.09*	0.07

Y = younger (< 50 years of age), O = older (50+ years of age),
SPD – statistical parity difference, DI = disparate index, T = Theil index

We have obtained the best HITL and AI Fairness Toolkit techniques and observed how they can help reduce age bias in CV screening. However, does our AI algorithm benefit from combining them, as other researchers have uncovered different biases? Also, does the order in which we apply these two techniques matter?

To examine the appropriate order, we examined the results where we applied the best HITL (pre-trained, single evaluator) treatment first, then fed them into the best AI Fairness Toolkit treatment (the combined pre-, in-, and post-processing). We then applied another treatment, using these two in reverse order. In each case, we began with the results from the original AI model to compare them to those treatments in which we used the HITL and AI Fairness Toolkit models in isolation. We hired eight additional crowdworkers to perform the HITL step on the 200 CVs in the first test set. We applied the results to the 200 candidate CVs in the second test set for each treatment to evaluate age bias removal effectiveness, just as we did for the previous treatments. The results are provided in Table 4.

TABLE IV. REDUCTION IN BIAS ON AGE BY ORDER OF TECHNIQUE APPLIED

Method Applied	Change in Selection Pct	SPD	DI	T
None	Y = 0.758, O = 0.525, **|Δ| = 0.000**	0.308	1.44	0.33
Best Model Using AI Fairness Toolkit Alone	Y = 0.654 O = 0.629, **|Δ| = 0.105**	*0.041*	*1.05*	0.05
Best Model Using HITL Approach Alone	Y = 0.679, O = 0.606, **|Δ| = 0.087**	*0.083*	*1.10*	0.08
Applying AI Fairness Toolkit first, then HITL	Y = 0.641, O = 0.637, **|Δ| = 0.004**	*0.034*	*1.01*	0.02
Applying HITL first, then AI Fairness Toolkit	Y = 0.642, O = 0.636, **|Δ| = 0.006**	*0.034*	*1.01*	0.02

A two-way ANOVA was performed to analyze the effect of the *model type* (combined vs. HITL only vs. AI Fairness Toolkit only) and *order* (HITL first vs. AI Fairness Toolkit first) on DI. It revealed no statistically significant interaction between the effects of the *model type* and *order* ($F(3, 196) = 10.12$, $p < 0.001$). Simple main effects analysis showed that the model type had a statistically significant effect on DI ($p < 0.001$). However, the order did not have a statistically significant impact on DI ($p < 0.6130$).

F. *Perception by Human Raters of Candidate Age*

A post-task questionnaire asked HITL raters if they knew the candidates' ages from the CV information. Raters indicated 69% age awareness despite our industry-standard methods of anonymizing this information. Crowdworkers reported an awareness of 58%, but our experts revealed an awareness of 77%, demonstrating how implicit biases in the language can potentially affect the decision-making process.

V. DISCUSSION

The previous section illustrates the challenges in mitigating age bias in the job selection process, even when applying state-of-the-art corrections. Our results complement those from earlier studies ([6], [7], [20], [23], [24]), although we evaluated using a different methodology and a different dataset and applying a different subset of techniques. Although the order in which we apply HITL and AI Fairness Toolkit models does not matter, applying both models in tandem gets us reasonably close to an unbiased model. These challenges can be addressed by shifting the perspective of older applicants, adjusting how algorithms handle information from older applicants, and encouraging more senior candidates to stay updated, particularly in fast-evolving fields like IT and healthcare.

Studies indicate that older workers are often perceived as less productive than their younger counterparts. While productivity measures may support this perception, their presence, especially in mentorship roles, can enhance the overall productivity of the workforce. Therefore, job descriptions that could benefit from mentor-mentee relationships should be rephrased to acknowledge the value older applicants bring. Including mentorship as a job attribute can level the playing field, recognizing that many older workers are well-qualified and eager to serve as mentors.

In addition, experts suggest that listing only the most recent 15-20 years of job experience can be beneficial, especially for applicants with a history of frequent job or role changes. Online job search platforms often scan resumes and CVs for ease of application, and algorithmically capping the job history window may minimize the potential downsides while highlighting valuable early career experiences. Similarly, not requiring the inclusion of dates for university degrees can divert attention from age-related determinants.

Job applicants can take various steps to enhance their selection prospects, particularly in fast-evolving fields like IT. Emphasizing up-to-date knowledge and using relevant buzzwords and jargon can be crucial, as AI filters frequently seek these indicators. Highlighting specific achievements, which most AI filters recognize, can also be advantageous. Furthermore, older job applicants often possess industry contacts and connections established over time, providing an advantage that younger applicants may not match.

While some argue for improved record-keeping to ensure equal opportunities for older applicants, it can complicate the already intricate hiring process. Employers may claim that such information was unknown at the time of candidate selection. However, the results from our study show that more than two-thirds can correctly place the candidates in the older/younger age group, even after attempts of anonymization are made. In job markets like the United States, where a third of the workforce is 50 or older, and this percentage is increasing, many employers must reevaluate their algorithmic hiring approaches. This evolving landscape requires a reassessment of current hiring methods.

Given the functioning of the AI Fairness toolkit, an expanded training dataset would lead to less significant reductions in fairness. The toolkit's effectiveness may be diminished in a real-world setting where a wider variety of CVs would be expected.

VI. CONCLUSION

AI stands poised to transform the hiring landscape, streamlining processes and offering data-driven insights into candidate qualifications. However, the integration of AI in hiring practices brings the risk of perpetuating and exacerbating existing biases, resulting in unfair decisions. One manifestation of AI bias is in resume screening tools, where algorithms may disadvantage certain candidates, favoring resumes with keywords associated with particular groups.

A significant contributor to AI bias is the reliance on biased training data, with systems learning and perpetuating existing biases based on age. Lack of transparency and explainability in AI hiring tools further compounds the issue, hindering efforts to identify and address bias effectively. Overreliance on AI can also lead to disregarding human judgment, emphasizing the importance of utilizing AI to complement human decision-making, not a replacement. In this paper, we have examined several strategies to mitigate this bias in the data. We reviewed the use of human-in-the-loop (HITL) techniques and the application of an open-source bias reduction toolkit technique, finding that combining the two methods provides the best approach to reducing the disparate impact of age groups on selection for job interviews. We found little benefit to using more than one human in the loop, nor from the order in which these two techniques are applied. Still, we did find an advantage from using pre-, in-, and post-processing methods for the AI Fairness Toolkit and in pre-training the humans used in the HITL technique. Therefore, establishing human review and override mechanisms, which HITL systems and AI Fairness Toolkits can provide, is important to reduce age-related biases.

Continuous monitoring and evaluation of AI systems are essential to detect and address biases proactively. Organizations should invest in AI Fairness Toolkits that prioritize transparency and explainability, enabling a clearer understanding of the decision-making processes. We have found that implementing a system incorporating HITL and AI Fairness Toolkits is the most effective approach to combatting age biases. By implementing these strategies, organizations can work towards mitigating AI unfairness in hiring and fostering a more equitable and inclusive recruitment process.

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